INCH-POUND

MIL-S-83502/1E
11 December 1989
SUPERSEDING
MIL-S-83502/1D
17 February 1988

MILITARY SPECIFICATION SHEET

SOCKETS, PLUG-IN ELECTRONIC COMPONENTS, ROUND STYLE (TO7, LOW PROFILE)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83502.

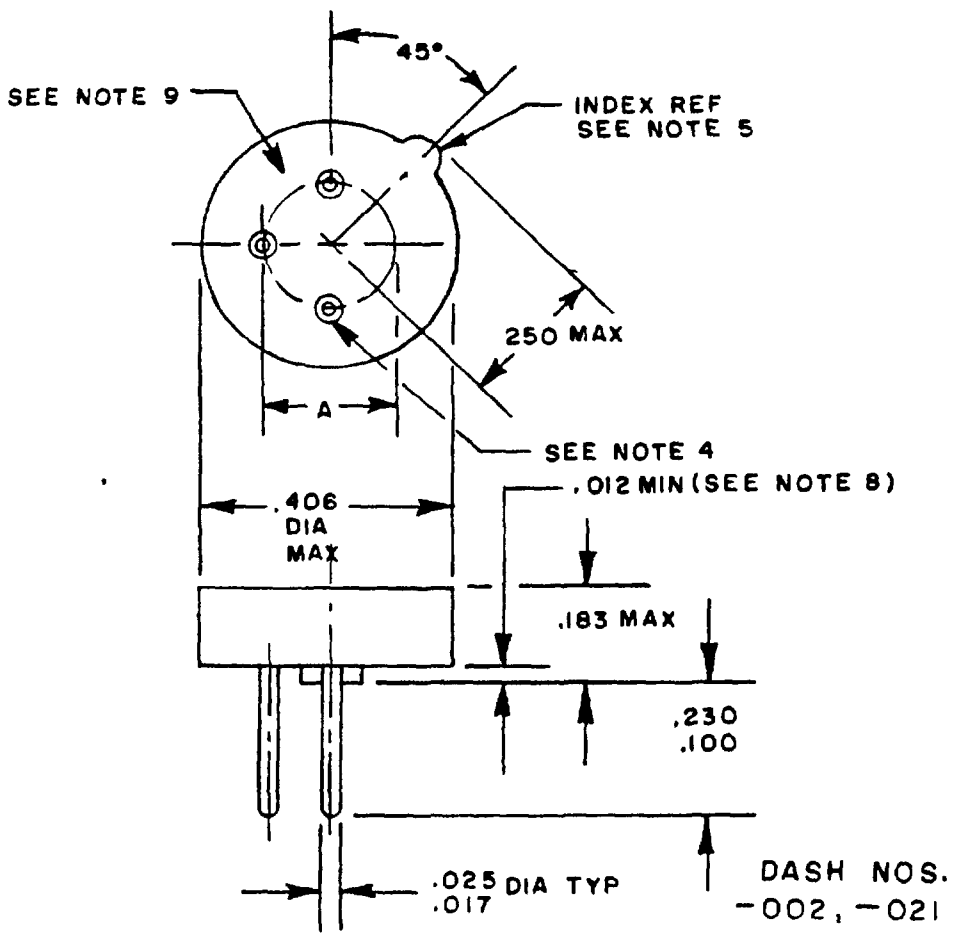
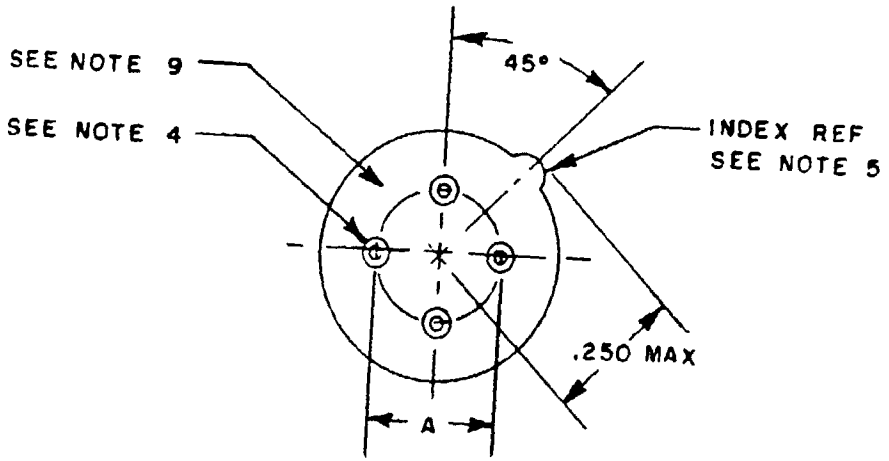


FIGURE 1. Dimensions and configurations.

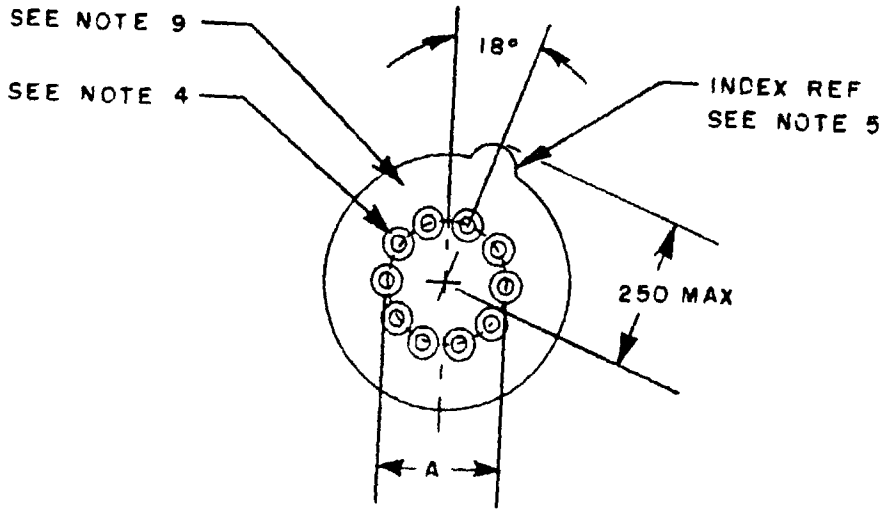
(E) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

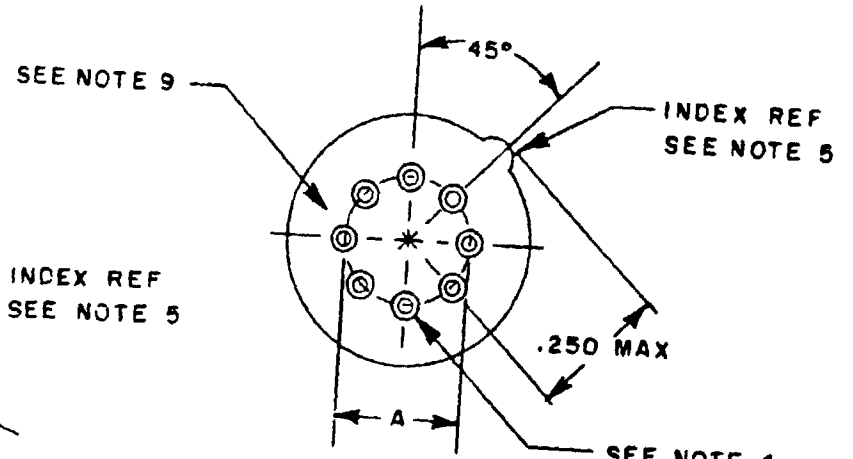
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DASH NOS.
-005, -024



DASH NOS
-017, -036



DASH NOS.
-011, -014, -030, -033

FIGURE 1. Dimensions and configurations - Continued.

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Applicable dash number	"A" Dimensions (pin circle)	
	Inches	mm
-002, -021	.200	5.08
-005, -024	.200	5.08
-011, -030	.200	5.08
-014, -033	.230	5.84
-017, -036	.230	5.84

Inches	mm
.001	0.25
.003	0.75
.017	0.43
.025	0.64
.039	0.98
.100	2.54
.183	4.65
.230	5.84
.250	6.35
.406	10.30

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. Device to accept leads, .017 inch (0.43 mm) diameter +.003, -.001 x .135 inch (3.43 mm)/ .160 inch long (4.06 mm). Contact shall be made at or before reaching a depth of .100 inch (2.54 mm)
5. A visual polarization index to be provided in the first quadrant. One method shown.
6. It is desirable that the socket face shall be relieved to facilitate removal of the integrated circuit.
7. Socket is intended for soldering to a PWB having plated through holes with a .039 inch (0.99 mm) diameter and a layout to suit the specific pin circle configuration.
8. Bosses (stand offs) shall be so designed as to prevent spacing between socket body and mounting board from being less than .012 inch (0.30 mm). This includes all dash numbers
9. Marking shall be on the package.

FIGURE 1 Dimensions and configurations - Continued.

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REQUIREMENTS.

Design and construction.

Dimensions and configuration: See figure 1.

Termination. Type II (printed circuit).

Material:

Ⓔ Body: Glass filled polyamide in accordance with ASTM D4000 PA110, diallyl phthalate, type SDG-F, in accordance with MIL-M-14, or TFE-fluorocarbon rod in accordance with ASTM D1710.

Contact: Beryllium copper in accordance with QQ-C-530 or C72900 spinodal alloy in accordance with ASTM B740.

Contact finish. In accordance with MIL-S-83502 (see table I).

Terminal Brass in accordance with QQ-B-626.

Finish: See table I.

Part or Identifying Number (PIN): M83502/1 -(dash number from table I).

TABLE I. Dash number and characteristics.

		Contact finish				
Dash number M83502/1-	Number of contacts	Engagement area	Termination and sleeve	Mating force (maximum) (lbs.)	Retention force (minimum) (lbs.)	Insulator color (body)
002	3	Gold	Gold	4.0	4.5	Red
005	4	Gold	Gold	4.0	4.5	Blue
011	8	Gold	Gold	4.0	4.5	Green
014	8	Gold	Gold	4.0	4.5	Orange
017	10	Gold	Gold	4.0	4.5	Yellow
021	3	Gold	Tin/lead	4.0	4 5	Red
024	4	Gold	Tin/lead	4.0	4.5	Blue
030	8	Gold	Tin/lead	4.0	4.5	Green
033	8	Gold	Tin/lead	2.5	3.0	Orange
036	10	Gold	Tin/lead	2.5	3.0	Yellow

These sockets are intended for use on .093 nominal printed circuit boards.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3724-01)

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